

BANARAS HINDU UNIVERSITY

B.A. / B.Sc. (Hons.) Semester II Mid-Term Test, 2024-25 (NEP)

DEPARTMENT OF MATHEMATICS**Paper Code: MATMJ-21 / MATMN-21 — Algebra**

Time: 1 Hour

Maximum Marks: 20

*Instructions: Attempt any **FOUR** questions. All questions carry equal marks (5 marks each).***Question 1**

- (a) Define n -th root of unity. Prove that if ω is an n -th root of unity, then $1 + \omega + \omega^2 + \dots + \omega^{n-1} = 0$. [3]
- (b) Let $z = 2 - 2i$. Find $|z|$, $\text{Arg}(z)$, $\arg z$, and $\sqrt{z}$. Plot z , $\arg z$, and $\sqrt{z}$ on the complex plane. [2]

Question 2

- (a) Let $p(x)$ be a polynomial with real coefficients. Prove that if $\alpha = x + iy$ is a root of $p(x)$ with $y \neq 0$, then $\bar{\alpha} = x - iy$ is also a root of $p(x)$. [3]
- (b) Find the smallest degree polynomial $p(x)$ with real coefficients which has roots 3 and $3 + 2i$. [2]

Question 3

- (a) Let a and b be two non-zero integers, and let $d = \gcd(a, b)$. Prove that there exist integers x and y such that $d = ax + by$. [3]
- (b) Using Euclidean algorithm, find $\gcd(8991, 12345)$. [2]

Question 4

- (a) Let $p(x) = ax^3 + bx^2 + cx + d$ be a cubic polynomial and let α, β, γ be its roots. Find the relation between the roots and coefficients of $p(x)$. [3]
- (b) Consider the polynomial $p(x) = x^3 - 5x^2 + 9x - 5$. If α, β, γ are the roots of $p(x)$, find the value of $\alpha + \beta + \gamma$, $+\beta\gamma + \gamma\alpha$, γ , and $\alpha^2 + \beta^2 + \gamma^2$. [2]

Question 5

- (a) Define equivalence relation. [2]
- (b) Let m be a fixed positive integer. Define a relation R on the set of integers $\mathbb{Z}$ such that $aRb \iff m \mid (a - b)$. Check whether R is an equivalence relation or not. If yes, for $m = 5$, find the equivalence class of 2. [3]